VNR VJIET

Name of the Laboratory:

Name of the Experiment:_____

Experiment No:_____ **Date:**_____

WEEK-8

AIM: Simulate frame allocation Methods. A) minimum number of frames B) equal allocation C) proportional allocation

a) minimum number of frames

ALGORITHM:

1. **Start**
2. Read the number of processes.
3. Read the total number of available frames.
4. Initialize `totalRequired = 0`.
5. For each process:
6. Read the minimum number of frames required by the process.
7. Add it to `totalRequired`.
8. Compare `totalRequired` with the total available frames.
9. If `totalRequired <= totalFrames`:
10. Display **"Minimum frame allocation is possible."**
11. Otherwise:
12. Display **"Minimum frame allocation is not possible."**
13. Display **"Not enough frames available."**
14. Display the total number of frames required.
15. **Stop**.

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Name of the Laboratory: _____

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Experiment No: _____ **Date:** _____

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Name of the Laboratory:

Name of the Experiment:_____

Experiment No:_____ **Date:**_____

PROGRAM:

```
#include <stdio.h>

int main()
{
   int processes, totalFrames;
   int i, minFrames, allocated = 0;

   printf("Enter number of processes: ");
   scanf("%d", &processes);

   printf("Enter total number of frames: ");
   scanf("%d", &totalFrames);

   printf("Enter minimum frames required by each process:\n");

   for (i = 0; i < processes; i++)
   {
      printf("Process %d: ", i + 1);
      scanf("%d", &minFrames);

      allocated += minFrames;
   }

   printf("\nTotal frames required = %d\n", allocated);

   if (allocated <= totalFrames)
   {
      printf("Minimum frame allocation is possible.\n");

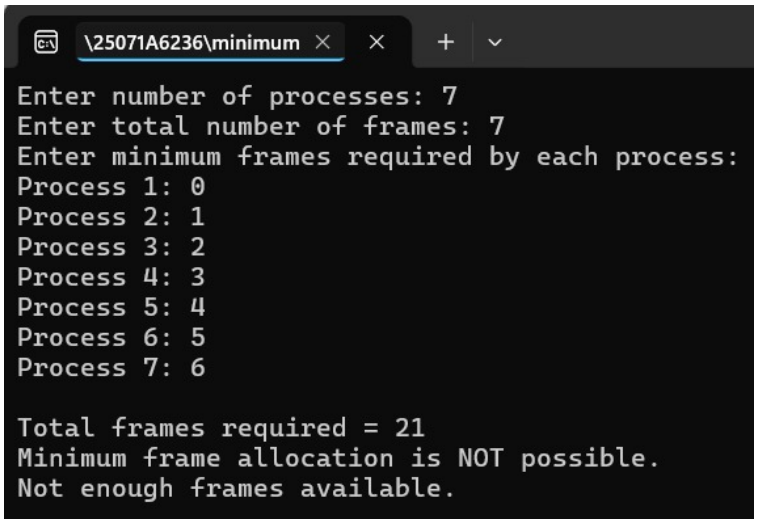
      printf("\nFrame Allocation:\n");

      for (i = 0; i < processes; i++)
      {
         printf("Process %d -> Minimum frames allocated\n", i + 1);
      }
   }
   else
   {
      printf("Minimum frame allocation is NOT possible.\n");
      printf("Not enough frames available.\n");
   }

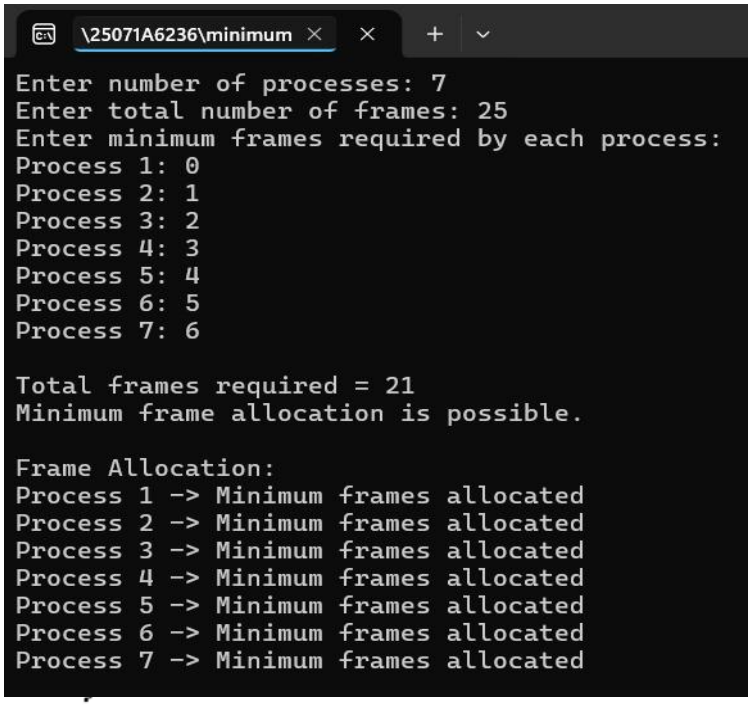
   return 0;
}
```

OUTPUT:

TEST CASE 1:



TEST CASE 2:



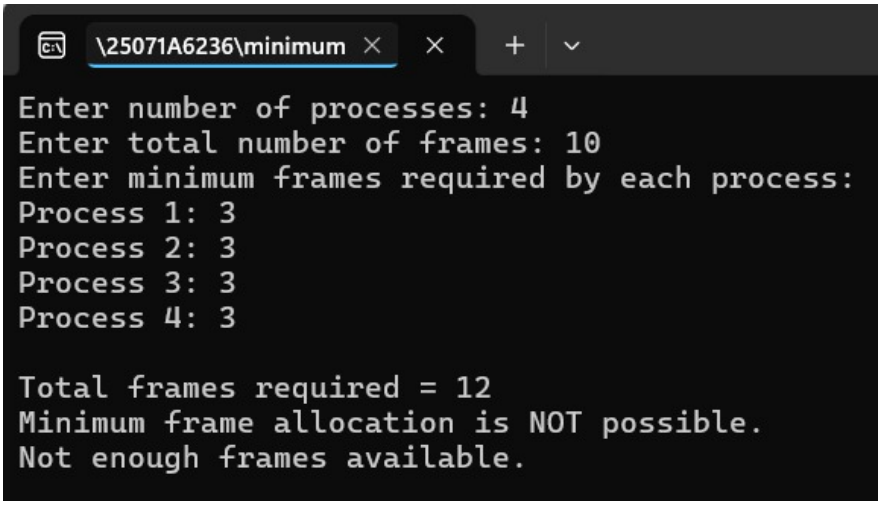
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Name of the Laboratory:

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TEST CASE 3:



B) EQUAL ALLOCATION

ALGORITHM:

1. **Start**
2. Read the number of processes `P`.
3. Read the total number of available frames `F`.
4. Calculate the number of frames allocated to each process:
5. `Allocation = F / P`
6. Calculate the remaining frames:
7. `Remaining = F % P`
8. For each process from 1 to `P`:
9. Allocate `Allocation` number of frames.
10. Display the frames allocated to the process.
11. Display the number of remaining frames.
12. **Stop**.

PROGRAM:

```
#include <stdio.h>
int main()
{
   int processes, frames;
   int allocation, remaining;
   int i;
   printf("Enter number of processes: ");
   scanf("%d", &processes);
   printf("Enter total number of frames: ");
   scanf("%d", &frames);

   allocation = frames / processes;
   remaining = frames % processes;
   printf("\nEqual Frame Allocation:\n");
   for (i = 1; i <= processes; i++)
   {
      printf("Process %d -> %d frames\n", i, allocation);
   }
   printf("\nFrames remaining = %d\n", remaining);

   return 0;
   }
```

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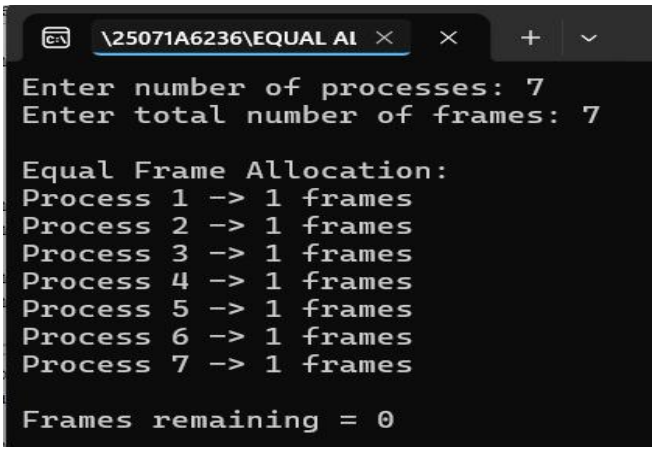
Name of the Laboratory:

Name of the Experiment:

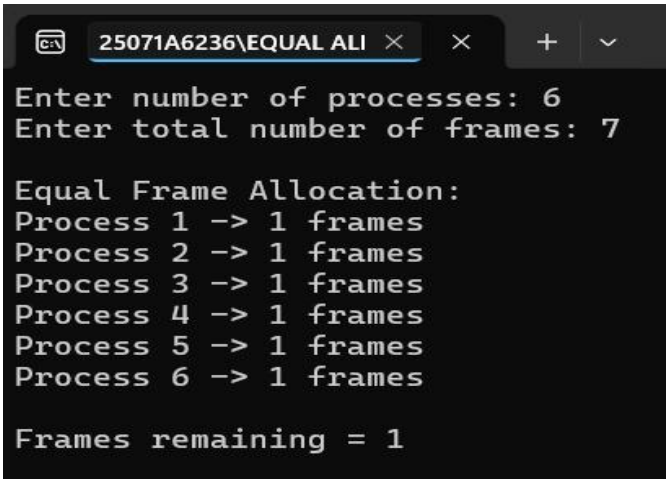
Experiment No: **Date:**

OUTPUT:

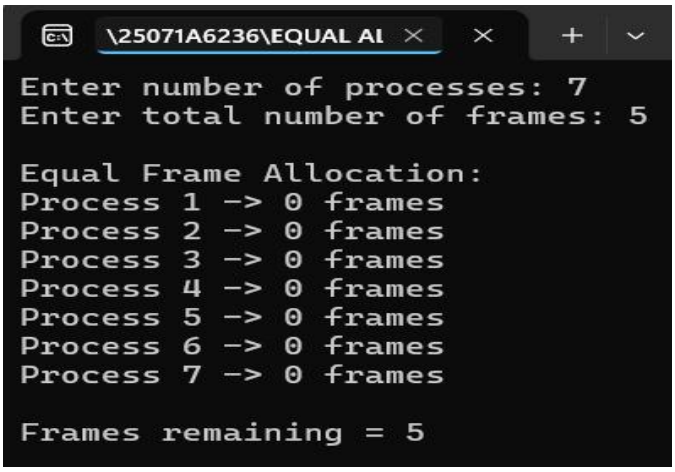
TEST CASE 1:



TEST CASE 2:



TEST CASE 3:



C) PROPORTIONAL ALLOCATION

ALGORITHM:

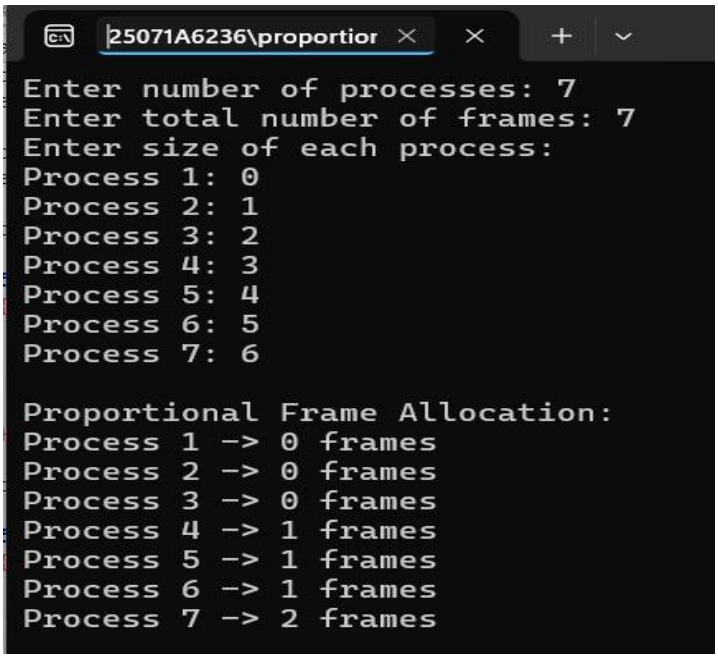
1. **Start**
2. Read the number of processes `N`.
3. Read the total number of available frames `F`.
4. Read the size of each process.
5. Calculate the total size of all processes:
6. `TotalSize = Size[1] + Size[2] + ... + Size[N]`
7. For each process, calculate the proportional number of frames:
8. `Frames[i] = (Size[i] × F) / TotalSize`
9. Display the number of frames allocated to each process.
10. **Stop**.

PROGRAM:

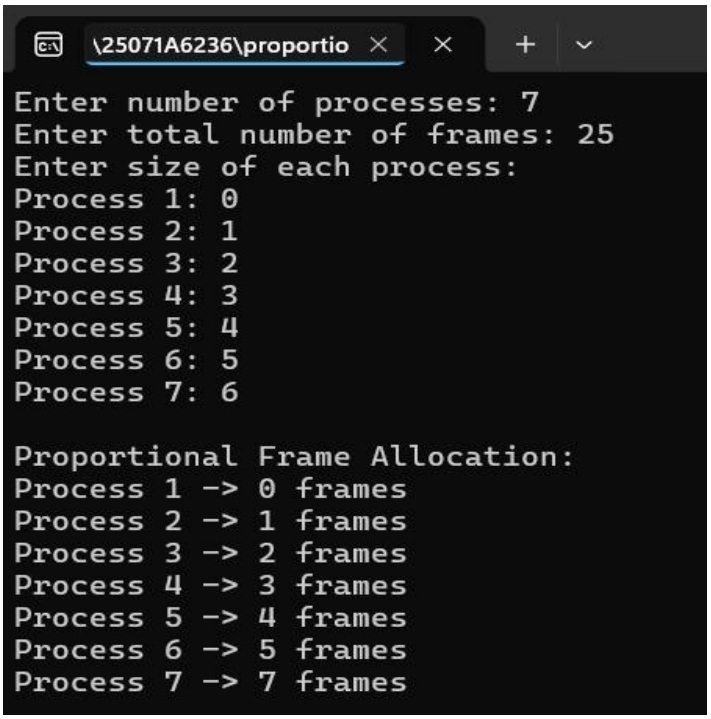
```
#include <stdio.h>
 int main()
{
 int n, totalFrames;
 int size[20], allocation[20];
 int totalSize = 0;
 int i;
 printf("Enter number of processes: ");
 scanf("%d", &n);
 printf("Enter total number of frames: ");
 scanf("%d", &totalFrames);
 printf("Enter size of each process:\n");
 for (i = 0; i < n; i++)
    {
 printf("Process %d: ", i + 1);
 scanf("%d", &size[i]);
 totalSize += size[i];
      }
 printf("\nProportional Frame Allocation:\n");
 for (i = 0; i < n; i++)
      {
      allocation[i] = (size[i] * totalFrames) / totalSize;
      printf("Process %d -> %d frames\n",
      i + 1, allocation[i]);
      }
  return 0;    }
```

OUTPUT:

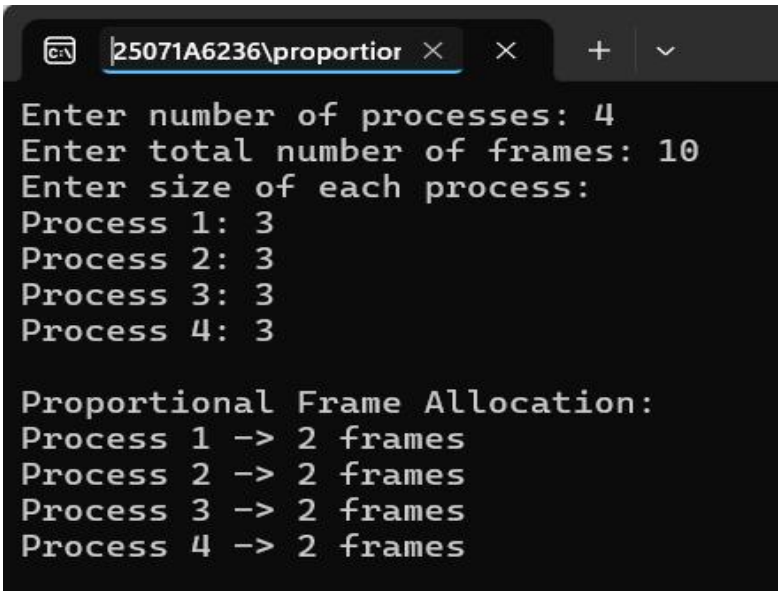
TEST CASE 1:



TEST CASE 2:



TEST CASE 3:


25071A6236\proportior
Enter number of processes: 4
Enter total number of frames: 10
Enter size of each process:
Process 1: 3
Process 2: 3
Process 3: 3
Process 4: 3
Proportional Frame Allocation:
Process 1 -> 2 frames
Process 2 -> 2 frames
Process 3 -> 2 frames
Process 4 -> 2 frames